

STAGE 1 - DATA ASSEMBLY

- Public sources only: World Bank prices, FAOSTAT, USDA PSD, FRED
- Target: year-over-year change in the deflated price, screened for a unit root
- Annual supply shifted to each producer's marketing-year start
- Country and region scales $\times$ production, yield, and sugar's cane and beet



STAGE 2 - MODEL COMPARISON

- Five families: random forest, boosting, CART, stepwise linear, elastic net
- Rolling origin: 35 test years on a 420-month window
- Every setting chosen on earlier test years only, never on the year scored
- Permutation importance in absolute error; the lowest-error family wins the cell



STAGE 3 - FORECASTING

- Three nested predictor sets by importance ratio, 50 / 20 / 10, floor of ten
- Fifteen names: nine learners, ETS, VAR, two ARIMAX, two naive
- Predictor set and hyperparameters chosen by the same honest rule
- One future-year forecast per case: 144 per crop, 720 in all



STAGE 4 - PUBLICATION

- Each forecaster re-scored with the year being judged left out
- Every case publishes the lowest-absolute-error model under its own name
- An 80 percent band from that model's own recorded errors
- State of the market and three volatility scenarios reported beside it



STAGE 5 - EXPLANATION

- The published model refit per fold, on the rows the forecast itself used
- Shapley values and LIME local surrogates for a released month
- Permutation importance and Friedman's H for pairwise interactions
- Four levels of detail, from an annual overview down to one month